

Yunmu Shu

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EDUCATION

Northeastern University <i>Master of Science in Computer Science (GPA: 3.9/4.0)</i>	Boston, MA Sep. 2022 – Aug. 2025
Northeastern University <i>Master of Science in Financial Mathematics</i>	Boston, MA Sep. 2021 – Aug. 2022
Guangdong University of Foreign Studies <i>Bachelor of Science in Economics</i>	Guangzhou, Guangdong Sep. 2017 – Jun. 2021

EXPERIENCE

Software Developer Intern <i>First Help Financial</i> - Engineered backend modules for an auto loan origination and decisioning platform used by dealers and underwriters, enhancing automation and ensuring regulatory compliance throughout the funding evaluation workflow - Analyzed slow SQL queries in loan transaction module using MySQL EXPLAIN, added composite indexes and optimized pagination with keyset query logic, reducing query latency by 22% under 10k+ records - Built a centralized Vehicle Data Retrieval service integrating Elasticsearch and Redis caching, boosting query throughput by 15% and handling 500k+ monthly search requests across multiple business modules - Monitored and visualized database performance metrics via Prometheus and Grafana to reduce overall CPU utilization	Jan. 2025 – Jun. 2025 Needham, MA
Software Developer Intern <i>Dassault Systèmes</i> - Contributed to backend development of ENOVIA document workflow platform within the 3DEXPERIENCE ecosystem, supporting regulatory compliance and enterprise collaboration - Architected and deployed role-based access control and audit logging for secure document approval and traceability - Built and optimized an API Gateway for multi-system document synchronization and metadata retrieval, integrating authentication and routing rules to improve throughput by 28% - Introduced centralized caching layer and API rate limiting to enhance system scalability and prevent redundant access - Developed asynchronous job handlers using Java Servlet and MQL to process large-scale document import tasks	Feb. 2024 – Jun. 2024 Waltham, MA
Software Developer Intern <i>Rapid Micro Biosystemes</i> - Enhanced backend architecture of a laboratory imaging platform for automated petri dish image capture and data storage - Optimized schema by restructuring entity mappings and introducing partitioning strategy to accelerate data retrieval by 25% - Refactored legacy PHP codebase into modular service components and integrated automated data migration scripts in Bash, ensuring consistency across 1M+ image records and improving maintainability and release stability - Containerized development environment with Docker and Apache to replicate production configuration	Jun. 2024 – Aug. 2024 Lexington, MA
Software Developer Intern <i>Tatfook Technology Co., Ltd</i> - Developed a responsive reporting dashboard for user-generated content analytics using Next.js, enabling real-time data visualization, trend tracking, and automated report generation for application users - Refined server-side rendering and API data fetching with Axios, reducing average page load time by 30% - Configured CI/CD pipelines utilizing GitLab for code uploads and Jenkins for seamless deployment	May. 2023 – Aug. 2023

PROJECTS

Tripadvisor App - Developed a Tripadvisor-like web application using Redis as middleware to handle high-concurrency scenarios - Engineered flash-sale module with read/write separation and atomic transaction logic through Redis Lua scripting - Integrated distributed Kafka message queue for asynchronous order processing, reducing API latency to tens of milliseconds - Designed user engagement features such as like/follow using ZSet and check-in tracking via Bitmap for efficient query	Oct. 2024 – Nov. 2024
GeeCache - Implemented a distributed caching system in Go, developing both local in-memory cache and HTTP-based cache with Least Recently Used strategy, and improved inter-node communication using protobuf - Employed Mutex locks to prevent cache breakdown and ensure data consistency under concurrent access - Integrated consistent hashing to select nodes, ensuring load balancing and efficient cache distribution	Jun. 2024 – Jul. 2024

TECHNICAL SKILLS

Programming Languages:	Java, C/C++, Go, TypeScript, JavaScript, SQL, Python
Frameworks:	Spring Boot, Spring Cloud, React.js, Next.js, Express.js, Node.js
Databases & Caching:	MySQL, Redis, MongoDB, PostgreSQL, DynamoDB, Kafka
Cloud & Tools:	AWS, Docker, Kubernetes, Jenkins, Nginx, Git, Terraform